



LinkedIn

Samuel Paul Lum

Mechanical Engineering Student
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EDUCATION

VIRGINIA POLYTECHNIC INSTITUTE AND STATE UNIVERSITY
Bachelor of Science in Mechanical Engineering
Expected May 2028

GRACE CHRISTIAN SCHOOL
Valedictorian | Staunton, VA
GPA: 4.32 | Graduated 2024

TECHNICAL SKILLS

CAD & Design | • Siemens NX • Fusion 360

Program & Analysis | • MATLAB • Statistical Analysis

Manufacturing & Fabrication | • CNC Machining • G-Code • Laser Cutting • 3D Printing • Welding • Sand Casting

Construction & Equipment | • Heavy Equipment Operation • Carpentry

CERTIFICATES

• Google Data Analytics Professional Certification • Google AI Essentials • Certified SOLIDWORKS Associate

EXPERIENCE

Westwood Professional Services | Staunton, Virginia

Survey/Civil Engineering Intern

Summer 2026

- Process total station and GPS survey data using Carlson Survey and AutoCAD Civil 3D.
- Draft survey linework, surface models, grading plans, and plats for residential and commercial land development projects.
- Perform field surveys, including as-built and improvement surveys, to collect and verify accurate site data.
- Collaborate with engineers and survey crews by balancing office drafting and data processing with field data collection.

Moffett Paving and Excavating | Staunton, Virginia

General Labor

May 2024-Aug 2025

- Supported residential and municipal paving, excavation, grading, and site preparation projects.
- Operated rollers and backhoes while assisting with grade verification using laser levels to support accurate earthwork.
- Utilized construction equipment and tools, including plate compactors, heavy-duty saws, and impact tools, while maintaining safety.
- Collaborated with experienced construction crews to complete projects efficiently in fast-paced field environments.

Virginia Tech Men's Soccer | Blacksburg, Virginia

Student Manager

Aug 2025 – present

- Coordinate equipment logistics for practices, competitions, and team travel.
- Manage setup, inventory, and transportation of training equipment to support daily team operations.
- Assist coaches with practice and game-day preparation for a Division I NCAA program.

PROJECTS

Smart Trash Bag Attachment (Virginia Tech Engineering Design Project)

Spring 2026

- Led the CAD design of a mechanical attachment intended to automate the sealing and removal of trash bags.
- Designed components in Siemens NX and prepared models for additive manufacturing using a Bambu Lab 3D printer.
- Fabricated and assembled a proof-of-concept prototype through multiple design iterations.
- Collaborated with a multidisciplinary engineering team to evaluate the design and identify opportunities for future improvement.

Smart Crosswalk Safety System (Virginia Tech Engineering Design Project)

Spring 2025

- Served as the primary CAD designer for a smart pedestrian crossing system intended to improve crosswalk safety.
- Designed structural components in Fusion 360 and assisted in constructing and testing the prototype.
- Integrated infrared sensors and ground lighting to demonstrate the system's safety concept.
- Collaborated with a multidisciplinary engineering team throughout design, fabrication, and testing.

LEADERSHIP & PROFESSIONAL INVOLVEMENT

American Society of Mechanical Engineers (ASME)

Member | 2025–Present

Chairman, National Honor Society | Grace Christian School

2023-2024

- Led chapter activities and coordinated community service totaling **400+ hours**